Chapter 1: Introduction to DBMS and the Relational Model

A database is an organised collection of data, and a Database Management System (DBMS) is the software that stores, retrieves and manages it. Compared with plain files, a DBMS reduces data redundancy, enforces integrity rules, supports multiple users at the same time and provides security and backup.

In the relational model, data is stored in tables called relations. Each row (tuple) is one record and each column (attribute) is one property of the record. A primary key is a column, or set of columns, that uniquely identifies each row and cannot be NULL. A foreign key is a column in one table that refers to the primary key of another table and is used to link tables together.

A candidate key is any minimal set of columns that could serve as a primary key. A superkey is any set of columns that uniquely identifies rows, even if it contains extra columns.

Chapter 2: SQL Basics

SQL (Structured Query Language) is the standard language for working with relational databases. It is divided into DDL (CREATE, ALTER, DROP), DML (SELECT, INSERT, UPDATE, DELETE), DCL (GRANT, REVOKE) and TCL (COMMIT, ROLLBACK).

The SELECT statement reads data. The WHERE clause filters rows, ORDER BY sorts the result (ascending by default, DESC for descending) and LIMIT restricts how many rows are returned. INSERT INTO adds a new row, UPDATE changes existing rows and DELETE removes rows. Always use a WHERE clause with UPDATE and DELETE, otherwise every row in the table is affected.

The difference between DELETE and TRUNCATE is that DELETE removes rows one by one and can be rolled back and filtered with WHERE, while TRUNCATE removes all rows at once and is faster. DROP TABLE removes the table structure itself.

```
SELECT name, marks
FROM students
WHERE marks > 70
ORDER BY marks DESC
LIMIT 5;
```

Chapter 3: Joins and Aggregation

A join combines rows from two or more tables based on a related column. An INNER JOIN returns only the rows that have matching values in both tables. A LEFT JOIN returns all rows from the left table and the matching rows from the right table, filling with NULL where there is no match. A RIGHT JOIN does the opposite, and a FULL OUTER JOIN returns all rows from both tables.

Aggregate functions calculate one value from many rows: COUNT, SUM, AVG, MIN and MAX. The GROUP BY clause groups rows that share a value so the aggregate is calculated per group. The HAVING clause filters groups after aggregation, while WHERE filters individual rows before grouping.

A subquery is a query nested inside another query. It can be used in the WHERE clause, for example to find students whose marks are above the average marks of the class.

```
SELECT d.name, COUNT(s.id) AS total
FROM departments d
LEFT JOIN students s ON s.dept_id = d.id
GROUP BY d.name
HAVING COUNT(s.id) > 10;
```

Chapter 4: Normalization

Normalization is the process of organising tables to reduce redundancy and avoid update, insert and delete anomalies. It works through a series of normal forms.

First Normal Form (1NF) requires that every column holds atomic (indivisible) values and that there are no repeating groups. Second Normal Form (2NF) requires 1NF and that every non-key column depends on the whole primary key, which removes partial dependencies and matters when the key is composite. Third Normal Form (3NF) requires 2NF and that non-key columns depend only on the key and not on other non-key columns, which removes transitive dependencies.

Boyce-Codd Normal Form (BCNF) is a stricter version of 3NF in which every determinant must be a candidate key. In practice, designers sometimes denormalize on purpose to make read-heavy queries faster, accepting some redundancy.

Chapter 5: Transactions and ACID Properties

A transaction is a group of database operations that must be treated as a single unit of work. A classic example is a bank transfer, which subtracts money from one account and adds it to another; either both steps succeed or neither does.

The ACID properties describe a reliable transaction. Atomicity means all operations happen or none do. Consistency means a transaction moves the database from one valid state to another valid state. Isolation means concurrent transactions do not interfere with each other. Durability means that once a transaction is committed, its changes survive a crash.

Concurrency problems include dirty reads (reading uncommitted data), non-repeatable reads and phantom reads. The database prevents them with isolation levels (READ UNCOMMITTED, READ COMMITTED, REPEATABLE READ and SERIALIZABLE) and with locking. Using COMMIT saves a transaction permanently and ROLLBACK undoes it.

Chapter 6: Indexing and Query Performance

An index is a separate data structure that speeds up searching on one or more columns, just like the index at the back of a book. Without an index the database must scan every row of the table, which is called a full table scan and takes O(n) time.

Most relational databases use a B-tree (or B+ tree) index, which keeps keys sorted and lets the database find a value in O(log n) time. A clustered index determines the physical order of the rows, so a table can have only one. Non-clustered indexes are stored separately and point to the rows. Hash indexes are very fast for equality lookups but cannot support range queries.

Indexes are not free: they use extra storage and make INSERT, UPDATE and DELETE slower because the index must be updated too. Index the columns that appear often in WHERE, JOIN and ORDER BY clauses, and use the EXPLAIN command to see how the database executes a query.